Exp No: 30

RTOS TASK SCHEDULING

Program

```
#include <systemc.h>
SC_MODULE(rtos_scheduler) {
void task1() {
while(true) {
cout << "Task1 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
void task2() {
while(true) {
cout << "Task2 Running at "
<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(rtos_scheduler) {
SC_THREAD(task1);
SC_THREAD(task2);
}
};
int sc_main(int argc, char* argv[]) {
rtos_scheduler obj("Scheduler");
sc_start(6, SC_SEC);
return 0;
}
```

Output

```
Compile done. Starting run...

        SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
        Copyright (c) 1996-2018 by all Contributors,
        ALL RIGHTS RESERVED
Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
Done
```